

KOKAR BOGSKAR, Finland

WMO: 029790

Lat: 59.504N

Lon: 20.347E

Elev: 16

StdP: 101.13

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.5	-6.4	-13.1	1.2	-7.3	-11.2	1.4	-5.7	20.1	-1.4	18.1	1.2	8.6	20	0.874

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.9	22.4	20.1	21.0	18.6	19.6	17.5	20.5	22.3	19.1	20.7	17.8	19.2	5.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	14.5	21.9	18.4	13.3	20.3	17.2	12.3	18.9	59.2	22.4	54.3	20.7	50.4	19.3	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.5	15.0	-9.8	22.6	3.3	2.1	-12.1	24.1	-14.1	25.4	-15.9	26.6	-18.3	28.1		
(5)			-10.2	20.3	3.2	1.7	-12.5	21.5	-14.4	22.5	-16.2	23.4	-18.5	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	0.2	-0.3	0.6	3.6	8.2	12.6	16.8	17.0	13.6	8.8	5.2	2.7	(6)
(7)		DBStd	6.65	3.29	3.21	2.85	2.07	2.69	2.29	2.48	1.96	2.17	2.46	2.69	2.88	(7)
(8)		HDD10.0	1572	304	288	292	192	69	4	0	0	2	52	143	226	(8)
(9)		HDD18.3	3987	563	521	550	442	313	173	63	49	142	294	393	484	(9)
(10)		CDD10.0	650	0	0	0	0	14	81	211	218	111	16	0	0	(10)
(11)		CDD18.3	24	0	0	0	0	0	0	15	9	0	0	0	0	(11)
(12)		CDH23.3	8	0	0	0	0	0	0	7	2	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.8	9.1	8.1	7.7	7.1	6.7	6.5	6.1	6.8	7.7	8.5	9.3	9.6	(14)
(15)	Precipitation	PrecAvg	831	83	60	52	39	40	59	67	87	74	90	94	87	(15)
(16)		PrecMax	1096	184	124	98	79	79	119	134	196	126	148	155	176	(16)
(17)		PrecMin	537	20	8	13	3	15	14	2	12	26	7	10	37	(17)
(18)		PrecStd	148	37	28	25	21	17	24	34	43	29	34	39	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	5.3	6.5	10.4	16.6	18.6	24.1	23.5	18.6	14.9	10.5	8.2	(19)
(20)			MCWB	5.5	3.8	4.3	6.3	12.2	14.8	20.3	21.3	16.8	13.8	9.5	7.2	(20)
(21)		2%	DB	5.7	4.4	5.4	8.4	14.5	17.1	22.9	22.0	17.6	13.6	9.7	7.4	(21)
(22)			MCWB	4.8	3.2	3.7	5.7	11.5	14.7	20.2	20.0	15.8	12.1	8.9	6.4	(22)
(23)		5%	DB	5.1	3.9	4.6	7.0	13.2	16.2	21.9	20.4	16.9	12.7	9.1	6.8	(23)
(24)			MCWB	4.3	3.1	3.3	5.1	10.5	14.2	19.6	18.0	15.2	11.5	8.2	5.8	(24)
(25)		10%	DB	4.4	3.4	3.8	6.1	11.8	15.4	20.5	19.5	16.3	12.0	8.5	6.1	(25)
(26)			MCWB	3.5	2.7	2.8	4.5	9.7	13.7	18.4	17.4	14.7	10.9	7.5	5.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	4.3	4.7	7.4	13.5	16.6	21.7	22.0	17.7	14.0	9.9	7.6	(27)
(28)			MCDB	6.2	4.8	5.8	9.6	15.4	17.8	22.8	23.2	18.5	14.8	10.1	8.1	(28)
(29)		2%	WB	5.1	3.7	4.1	6.2	12.2	15.6	20.9	20.3	16.2	12.5	9.1	6.7	(29)
(30)			MCDB	5.6	4.1	5.0	7.8	13.9	16.6	22.6	21.7	17.2	13.2	9.6	7.2	(30)
(31)		5%	WB	4.5	3.2	3.5	5.4	11.0	14.8	19.9	18.7	15.6	11.9	8.4	6.1	(31)
(32)			MCDB	5.0	3.7	4.4	6.8	12.7	15.8	21.9	20.0	16.6	12.6	9.0	6.7	(32)
(33)		10%	WB	3.6	2.8	2.9	4.7	10.1	14.0	18.5	17.8	15.1	11.2	7.7	5.3	(33)
(34)			MCDB	4.4	3.4	3.7	5.8	11.6	15.2	20.4	19.1	16.1	11.9	8.4	6.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	2.6	2.4	2.4	2.5	2.7	2.5	2.1	1.9	2.0	2.1	2.2	2.4	(35)
(36)			MCDBR	2.6	2.4	3.1	4.0	4.8	3.5	2.8	2.1	2.1	2.3	2.1	2.6	(36)
(37)			MCWBR	2.8	2.2	2.2	2.4	2.9	2.5	2.9	2.3	2.3	2.4	2.4	2.8	(37)
(38)		5% WB	MCDBR	2.6	2.4	3.0	3.9	4.3	3.0	2.5	1.9	1.9	2.4	2.0	2.4	(38)
(39)			MCWBR	2.9	2.4	2.3	2.6	3.1	2.6	2.8	2.3	2.3	2.6	2.3	2.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.320	0.337	0.373	0.368	0.361	0.390	0.381	0.350	0.337	0.313	0.271	(40)	
(41)		taud	2.217	2.297	2.347	2.343	2.433	2.492	2.434	2.442	2.509	2.410	2.233	2.128	(41)	
(42)		Ebn at Noon	503	687	798	828	862	873	835	814	784	664	457	367	(42)	
(43)		Edh at Noon	43	68	87	104	102	98	102	94	75	61	43	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.26	0.80	2.36	4.16	5.71	6.26	5.86	4.59	2.87	1.22	0.35	0.17	(44)	
(45)		RadStd	0.03	0.14	0.26	0.43	0.51	0.37	0.47	0.38	0.33	0.18	0.08	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.34	N/A	N/A	N/A	N/A	N/A	-105	N/A	N/A	N/A

Nomenclature: See separate page